

# Security Linter Report

|                     |                           |
|---------------------|---------------------------|
| Filename            | case10_dynamic_columns.py |
| Timestamp           | 2026-06-05 05:27:59 UTC   |
| Lines of Code (LOC) | 25                        |
| Verdict             | <b>VULNERABLE</b>         |

| ID         | Severity | Sink           | Source Variable | Source Type | Line |
|------------|----------|----------------|-----------------|-------------|------|
| CVE-SL-001 | CRITICAL | cursor.execute | sql             | STDIN       | 33   |
| CVE-SL-002 | CRITICAL | cursor.execute | table_name      | STDIN       | 34   |

|          |   |
|----------|---|
| CRITICAL | 2 |
| WARNING  | 0 |
| INFO     | 0 |
| TOTAL    | 2 |

**CVE-SL-001 — cursor.execute****CRITICAL**

|                              |                              |
|------------------------------|------------------------------|
| <b>CWE</b>                   | CWE-89                       |
| <b>Sink</b>                  | cursor.execute               |
| <b>Sink Line</b>             | 33                           |
| <b>Sink Column</b>           | 0                            |
| <b>Sink Expression</b>       | cursor.execute(sql)          |
| <b>Argument</b>              | sql                          |
| <b>Source Variable</b>       | sql                          |
| <b>Source Type</b>           | STDIN                        |
| <b>Propagation Mechanism</b> | direct assignment from stdin |

***Taint Path Trace:***

| Step | Role   | Line | Expression                                                      |
|------|--------|------|-----------------------------------------------------------------|
| 1    | SOURCE | 10   | sql = "SELECT " + column_name + " FROM " + table_name + " WH... |
| 2    | SINK   | 33   | cursor.execute(sql)                                             |

***Remediation:***

Use a parameterized query and validate the input type. If the value must be an integer, cast it with int():

```
user_id = int(input("ID: "))
query = "SELECT * FROM users WHERE id = ?"
cursor.execute(query, (user_id,))
```

**CVE-SL-002 — cursor.execute****CRITICAL**

|                              |                              |
|------------------------------|------------------------------|
| <b>CWE</b>                   | CWE-89                       |
| <b>Sink</b>                  | cursor.execute               |
| <b>Sink Line</b>             | 34                           |
| <b>Sink Column</b>           | 0                            |
| <b>Sink Expression</b>       | cursor.execute(base_sql)     |
| <b>Argument</b>              | base_sql                     |
| <b>Source Variable</b>       | table_name                   |
| <b>Source Type</b>           | STDIN                        |
| <b>Propagation Mechanism</b> | direct assignment from stdin |

### ***Taint Path Trace:***

| Step | Role   | Line | Expression                                     |
|------|--------|------|------------------------------------------------|
| 1    | SOURCE | 6    | table_name = input("Enter table name: ")       |
| 2    | ASSIGN | 18   | base_sql += " AND category = " + category + "" |
| 3    | SINK   | 34   | cursor.execute(base_sql)                       |

### ***Remediation:***

Use a parameterized query and validate the input type. If the value must be an integer, cast it with int():

```
user_id = int(input("ID: "))
query = "SELECT * FROM users WHERE id = ?"
cursor.execute(query, (user_id,))
```

---

No safe sink paths confirmed.

No sanitization functions detected.

|                                          |                |
|------------------------------------------|----------------|
| CFG — Total Nodes                        | 20             |
| CFG — Condition/Loop Nodes               | 0              |
| CFG — Assignment Nodes                   | 13             |
| CFG — Call Nodes                         | 5              |
| CFG — Merge Nodes                        | 0              |
| ████████████████████████████████████████ | ██████████████ |
| DFG — Total Nodes                        | 63             |
| DFG — Total Edges                        | 65             |
| DFG — Source Nodes                       | 7              |
| DFG — Sink Nodes                         | 2              |
| DFG — Operator Nodes                     | 15             |
| DFG — Constant Nodes                     | 18             |
| DFG — Variable Nodes                     | 18             |
| Taint Propagation Hops                   | 17             |
| ████████████████████████████████████████ | ██████████████ |
| Analysis Time (ms)                       | 0.3            |

|                        |                                 |
|------------------------|---------------------------------|
| <b>Vulnerabilities</b> | 2 CRITICAL   0 WARNING   0 INFO |
| <b>Safe Sinks</b>      | 0                               |
| <b>Analysis Time</b>   | 0.3 ms                          |
| <b>Engine Version</b>  | 1.0.0                           |
| <b>Final Verdict</b>   | <b>VULNERABLE</b>               |